AMENDMENT OF SOLICITATION/MODIFICATION OF CONTRACT

1. CONTRACT ID CODE	PAGE OF PAGES
	1 of 8

2. AMENDMENT/MODIFICATION NO.	3. EFFECTIVE DATE	4. REQUISITION/PURCHASE REQ. NO.	5. PROJECT NO. (If applicable)
0002	18-Jul-2025	0012296018-0001	

6. ISSUED BY CODE W911W6	7. ADMINISTERED BY (If other than item 6) CODE
ARMY CONTRACTING COMMAND (ACC) REDSTONE/CCAM-MZD-B BLDG 401 VICTORY BLVD FORT EUSTIS VA 23604-5577	**See Item 6**

8. NAME AND ADDRESS OF CONTRACTOR (No., Street, County, State and Zip Code)		
	X	9A. AMENDMENT OF SOLICITATION NO. W911W625R0006
	X	9B. DATED (SEE ITEM 11) 03-Jul-2025
		10A. MOD. OF CONTRACT/ORDER NO.
CODE / FACILITY CODE		10B. DATED (SEE ITEM 13)

11. THIS ITEM ONLY APPLIES TO AMENDMENTS OF SOLICITATIONS

[X] The above numbered solicitation is amended as set forth in Item 14. The hour and date specified for receipt of Offer [ ] is extended, [X] is not extended.

Offer must acknowledge receipt of this amendment prior to the hour and date specified in the solicitation or as amended by one of the following methods:
(a) By completing Items 8 and 15, and returning ______ copies of the amendment; (b) By acknowledging receipt of this amendment on each copy of the offer submitted; or (c) By separate letter or telegram which includes a reference to the solicitation and amendment numbers. FAILURE OF YOUR ACKNOWLEDGMENT TO BE RECEIVED AT THE PLACE DESIGNATED FOR THE RECEIPT OF OFFERS PRIOR TO THE HOUR AND DATE SPECIFIED MAY RESULT IN REJECTION OF YOUR OFFER. If by virtue of this amendment you desire to change an offer already submitted, such change may be made by telegram or letter, provided each telegram or letter makes reference to the solicitation and this amendment, and is received prior to the opening hour and date specified.

12. ACCOUNTING AND APPROPRIATION DATA (If required)

13. THIS ITEM APPLIES ONLY TO MODIFICATIONS OF CONTRACTS/ORDERS. IT MODIFIES THE CONTRACT/ORDER NO. AS DESCRIBED IN ITEM 14.

[ ] A. THIS CHANGE ORDER IS ISSUED PURSUANT TO: (Specify authority) THE CHANGES SET FORTH IN ITEM 14 ARE MADE IN THE CONTRACT ORDER NO. IN ITEM 10A.

[ ] B. THE ABOVE NUMBERED CONTRACT/ORDER IS MODIFIED TO REFLECT THE ADMINISTRATIVE CHANGES (such as changes in paying office, appropriation date, etc.) SET FORTH IN ITEM 14, PURSUANT TO THE AUTHORITY OF FAR 43.103(B).

[ ] C. THIS SUPPLEMENTAL AGREEMENT IS ENTERED INTO PURSUANT TO AUTHORITY OF:

[ ] D. OTHER (Specify type of modification and authority)

E. IMPORTANT: Contractor [ ] is not, [ ] is required to sign this document and return ______ copies to the issuing office.

14. DESCRIPTION OF AMENDMENT/MODIFICATION (Organized by UCF section headings, including solicitation/contract subject matter where feasible.)

See page 2.

Except as provided herein, all terms and conditions of the document referenced in Item 9A or 10A, as heretofore changed, remains unchanged and in full force and effect.

15A. NAME AND TITLE OF SIGNER (Type or print)		16A. NAME AND TITLE OF CONTRACTING OFFICER (Type or print) TEL: EMAIL:	
15B. CONTRACTOR/OFFEROR (Signature of person authorized to sign)	15C. DATE SIGNED	16B. UNITED STATES OF AMERICA BY (Signature of Contracting Officer)	16C. DATE SIGNED 18-Jul-2025

SECTION SF 30 BLOCK 14 CONTINUATION PAGE

SUMMARY OF CHANGES

SECTION SF 30 - BLOCK 14 CONTINUATION PAGE

The following have been added by full text:

AMD 0002

1. The above numbered Broad Agency Announcement (BAA) is hereby modified to:
    a. Clarify that AMENDMENT 0001 was not published and to clarify that **AMENDMENT 0002 is the first amendment applicable to the BAA**.

    b. Issue Attachment 5, Sequence #25-ATTACH5-A and Sequence #25-ATTACH5-B which provides descriptions of the following Topics:

       25-ATTACH5-A
       1—Topological Optimization for Additively Manufactured (AM) Corrosion Resistant Gearbox Housings
       2--Lightweight Emergency Loss of Lubrication System
       3—Artificial Intelligence and Machine Learning for Drive System Prognostics
       4—Non-Press Fit or Splined Inner/Outer Race Fully Ceramic Bearings in Gearboxes
       5--Topological Optimization of Dynamic Drive System Components
       25-ATTACH5-B
       1—Hybrid VTOL to Enable Operational Capability
    c. This confirms that the two documents published July 3, 2025, in SAM.gov entitled "Aviation and Missile R&D BAA" and "ATTACHMENT 5 Notice of Funding Opportunity to BAA W911W6-25-R-0006" are the same document which is the Broad Agency Announcement (W911W6-25-R-0006); and neither published document contains Attachment 5.

    d. For topics which are open continuous, offerors are requested to provide concept papers for one or more of the listed topics of interest before 1 October of each year. Concept papers will be reviewed in accordance with Aviation and Missile R&D BAA W911W6-25-R-0006 Attachment 2, Two Step Process with requests for full proposals to follow, as applicable.

SECTION L - INSTRUCTIONS, CONDITIONS AND NOTICES TO BIDDERS

The following have been added by full text:

ATTACH 5 TOPICS AREAS OF INT

Attachment 5 to BAA W911W6-25-R-0006
Topics/Areas of Interest

Sequence #25-ATTACH5-A
Supplemental Power, Efficient Engines, and Drives (SPEED)

Description: The Technology Development Directorate (TDD) is interested in the development and demonstration of rotorcraft engine and drive system technologies as part of the "Supplemental Power, Efficient Engines, and Drives (SPEED)" program. Specific objectives related to rotorcraft engines include specific fuel consumption improvements, power density improvements, sustainability improvements, and surge margin improvements. Specific objectives related to rotorcraft drive systems include component life improvements, power density improvements, life cycle cost reductions, and loss of lubrication performance improvements. Specific topics of interest include Topological Optimization for Additively Manufactured Corrosion Resistant Gearbox Housings, Lightweight Emergency Loss of Lubrication Systems, Artificial Intelligence and Machine Learning for Drive System Prognostics, Non-Press Fit or Splined Inner/Outer Race Fully Ceramic Bearings in Gearboxes, and Topological Optimization of Dynamic Drive System Components.

Topic Audience: Small Businesses, Large Businesses

Department of the Army Research, Development, Test, & Evaluation (RDT&E) Budget Activity (BA): BA 2, Applied Research. program that is seeking technologies starting between a Department of Defense (DoD) Technology Readiness Level (TRL) 3 to 4 and ending at a TRL 5. Sources not familiar with DoD TRLs, please review the DoD Technology Readiness Assessment Guidebook published by the Under Secretary of Defense for Research and Engineering.

Program Element: PE 0602183A / *Air Platform Applied Research*

Project: DK1 / *Air Vehicle Integrated and Alternative Tech (AVIATe)*

Technology Areas: Engines, Drive Systems

Budget:

FY26	FY27	FY28	FY29	FY30	Total
$1,500K	$1,900K	$2,700K	$2,600K	$2,700K	$11,400K

Schedule: Multiple awards are anticipated each year, based upon availability of funds. The Government anticipates that it will begin selection of concept papers during the first quarter of each fiscal year. Therefore, offerors submitting concept papers before 1 October are better positioned for consideration. Each award is anticipated to have a period of performance between 12-24 months.

Alterations from Standard Process: N/A

Opportunities for Fundamental Research: No. This work will require use of controlled unclassified information (CUI) and generation of CUI.

Technical POC: Matthew Clark (757-878-2400, matthew.b.clark34.civ@army.mil)

1--Topic Name: Topological Optimization for Additively Manufactured (AM) Corrosion Resistant Gearbox Housings

Preferred Research Focus: Manned and Unmanned Army Rotorcraft Drive Systems

Basis of Need/Capabilities Gap: Rotorcraft gearbox housings are typically fabricated using a Magnesium alloy and produced by either a casting or "hogout" process. Coatings are often used to prevent corrosion in Magnesium housings, but corrosion may still occur when the coatings are damaged by foreign objects. Aluminum is a potential alternate material, but galvanic corrosion can occur due to the use of steel bolts. The casting and hogout manufacturing processes often require increased thickness and create material waste during fabrication. The casting process also often requires several iterations to fabricate an acceptable new gearbox housing design and often causes material flaws which leads to scrapped housings.

Solution Guidelines: TDD seeks technologies related to topological optimization of additively manufactured corrosion resistant gearbox housings. Challenges that must be overcome in this area include reducing housing weight while maintaining performance requirements, reducing post processing requirements, detecting and correcting various fault types during the AM process, and preventing housing corrosion with a lightweight solution that is not easily damaged. Efforts are expected to perform component testing to demonstrate the challenges have been addressed and provide an analysis of the benefits towards an Army rotorcraft.

Anticipated benefits: Anticipated benefits for this topic include improved power density (due to the reduction in weight from topological optimization), increased component life (due to the reduction in corrosion and material flaws), and reduced life cycle cost (due to the reduction in corrosion, scrap rate, design iteration time, and required material).

2--Topic Name: Lightweight Emergency Loss of Lubrication System

Preferred Research Focus: Manned and Unmanned Army Rotorcraft Drive Systems

Basis of Need/Capabilities Gap: Army rotorcraft gearboxes are required to operate for 30 minutes after a loss of lubrication (LOL) event to provide sufficient time to safely land. The Army desires increased LOL emergency operation time to allow more time for pilot decision making on how and where to land, and to escape hostile territory before landing. Gear coatings can potentially provide a lightweight solution to improve LOL capability but have not yet demonstrated durability over time and may not be a sufficient solution for all configurations.

Solution Guidelines: TDD seeks technologies which improve the LOL capability of rotorcraft gearboxes, particularly using small, pressurized containers of emergency lubrication material that is expelled upon temperature rise in the gearbox. Challenges to overcome include the ability to ensure lubricant release at set elevated temperature conditions, device reliability independent of manual inspection, and optimizing the amount and location of devices for reduced weight and cost while maintaining effectiveness. Efforts are expected to perform component testing to demonstrate the challenges have been addressed as well as small scale testing in a demonstrator gearbox to assess performance.

Anticipated Benefits: Anticipated benefits for this topic include improved power density (due to reduced LOL system weight) and increased LOL time (due to improvements to the emergency LOL system).

3--Topic Name: Artificial Intelligence and Machine Learning for Drive System Prognostics

Preferred Research Focus: Manned and Unmanned Army Rotorcraft Drive Systems

Basis of Need/Capabilities Gap: Operation and Sustainment costs account for roughly 70% of the total life cycle cost of rotorcraft, with maintenance of drive systems being one of the top cost drivers. High maintenance costs are often driven by early removal of components well before failure. Condition indicators from current generation health management systems enable some capability to predict failure but often require decades of usage data for refinement into a useful tool for maintenance forecasting. Future Vertical Lift (FVL) rotorcraft will not have access to past failure data to initially develop condition indicators, leading to overly conservative inspection intervals for drive components early in the aircraft life cycle. Artificial Intelligence (AI) models are needed to predict drive system failures in FVL rotorcraft and eliminate overly conservative inspection intervals.

Solution Guidelines: TDD seeks the development of AI models that can integrate physics-based models with actual usage data to accurately predict failures for new aircraft. Challenges to overcome include determining the appropriate AI algorithm, collecting sufficient amounts and types of data for model training, determining appropriate sensor and usage data required, and validating accuracy of failure forecasting. Efforts are expected to combine physics-based and AI models and demonstrate their failure forecasting accuracy against a dataset of known failures.

Anticipated Benefits: Anticipated benefits for this topic include reduced life cycle cost (due to accelerated maturity of next-generation condition indicators) and increased aircraft availability (due to elimination of conservative maintenance intervals).

4--Topic Name: Non-Press Fit or Splined Inner/Outer Race Fully Ceramic Bearings in Gearboxes

Preferred Research Focus: Manned and Unmanned Army Rotorcraft Drive Systems

Basis of Need/Capabilities Gap: Fully ceramic bearings provide up to a 50% weight reduction compared to steel bearings, but the mismatch in coefficient of thermal expansion (CTE) between ceramic bearings and steel gear shafts causes cracking in the bearing inner ring. Tolerance rings have shown to allow some thermal mismatch expansion but are not designed for shaft side loading common in aviation transmissions. This side loading causes severe compression of the tolerance rings, resulting in bearing and shaft misalignment and ultimately reduced reliability and component life.

Solution Guidelines: TDD seeks a novel bearing design incorporating a spliced inner and/or outer race is needed to accommodate CTE mismatch and side loading. Challenges to overcome include maintaining required bearing tolerances, eliminating the formation of stress risers at the spline root, and identifying a method to detect non-metallic debris in the oil. Efforts are expected to perform component thermal testing to demonstrate performance.

Anticipated Benefits: Anticipated benefits for this topic include increased power density (due to reduced bearing weight) and increased LOL time and component life (due to improved durability of ceramics).

5--Topic Name: Topological Optimization of Dynamic Drive System Components

Preferred Research Focus: Manned and Unmanned Army Rotorcraft Drive Systems

Basis of Need/Capabilities Gap: Dynamic drive system component design can be limited by traditional forging and machining processes. For example, traditional processes may require bolting or welding components together. These traditional processes may impact manufacturing time and cost, as well as system weight.

Solution Guidelines: TDD seeks topological optimization with new manufacturing processes such as additive manufacturing, which enables more complex geometries as well as reductions in cost and weight. Challenges to overcome include developing an optimized design that reduces weight while maintaining strength, and determining effective manufacturing methods for the topologically optimized designs. Efforts are expected to perform comparison testing between topologically optimized and traditionally manufactured components to assess weight, cost, and performance impacts.

Anticipated Benefits: Anticipated benefits for this topic include increased power density (due to reduced weight) and reduced life cycle cost (due to the reduction of material required and increased manufacturing speed).

Sequence #25-ATTACH5-B
Hybrid VTOL to Enable Operational Capabilities

Description: The U.S. Army is soliciting critical data and insights on hybrid vertical take-off and landing (VTOL) and short take-off and landing (STOL) aircraft concepts. The Army is seeking to leverage industry expertise to inform Army decision-making regarding the feasibility, operational impact, cost of adoption, and requirements applicable to hybrid technologies for future Army missions.

Topic Audience: Small Business, Large Business, Academia, HBCU/MI

Department of the Army Research, Development, Test, & Evaluation (RDT&E) Budget Activity (BA): 2, Applied Research

Program Element(s): PE 0602183A / Air Platform Applied Research

Project: DK1 / Air Vehicle Integrated & Alternative Tech (AVIATe)

Discipline(s): Hybrid aviation systems, Vertical/short take-off and landing, Unmanned aircraft

Budget:

FY25	FY26	FY27	Total
$400K	$1,300K	$1,500	$3,200K

Approximate planning numbers

Multiple awards are anticipated, based upon availability of funds. Two to four awards are anticipated with a period of performance between 12-24 months.

Alterations from Standard Process: N/A

Program Schedule: Project schedule is anticipated to conclude at the end of FY27.

Opportunities for Fundamental Research No.

Government Technical POC: David Friedmann, 757-878-2159, david.s.friedmann.civ@army.mil

1--Topic Name: Hybrid VTOL to Enable Operational Capability

Preferred Research Focus: The Government seeks data to provide insights into the technical feasibility, operational impact, and cost implications of integrated hybrid architectures, technologies, and system designs for vertical take-off and landing (VTOL) and very short take-off and landing (STOL) aircraft platforms configured for relevant Army missions (it is assumed that applicants will have knowledge/ideas in this area). Of particular interest:

- Performance enhancements (range, endurance, payload), efficiencies, novel modes of operation (e.g. quiet approach) or other beneficial attributes enabled by hybrid technologies.
- Cost and operational impact.
- Technical challenges and limitations; physical constraints, maturity and development needs; failure modes and mitigations; integration issues and challenges; benchmark weight, efficiencies, and performance of integrated hybrid power implementation schemes.
- Degree of specialized militarization considerations required for suitable performance of relevant Army missions.

Basis of Need/Capabilities Gap: The Army seeks to leverage on-going Industry investments to assess the unique characteristics of hybrid systems that potentially provide warfighters the ability to conduct operations more efficiently and effectively, or enable new modes of operation that enhance warfighter outcomes. This effort is intended to identify ways in which a hybrid VTOL/STOL solution could bring added value to Army operations.

Solution Guidelines: Provide data, information, and insights into the technology building blocks and costs of hybrid aircraft for Army relevant use cases. Develop and validate study insights through analysis, modeling, test data, and demonstrations. Further guidelines are provided below.

- Information and data are desired to inform further risk mitigation, technology maturation planning, program planning, requirements development, and to inform stakeholders inside and outside of the Government of significant findings and developments.
- Unoccupied aircraft of Group 4/5-size (up to approximately 10,000 lbs. at take-off) with vertical take-off and landing (VTOL) and very short take-off and landing (STOL) aircraft are of primary interest. Data from surrogates may be acceptable if it is directly applicable to aircraft described herein.
- Notional use cases and sizing points range from simple low-cost supply/logistics missions in mostly benign conditions to various tactical missions within a threat environment as a means of extending reach (in time and distance) of onboard or deployable effects.
- The scope of this initial Government-funded effort is anticipated to be primarily analytical. However, it may include system integration and testing depending on individual entry points based on maturity of applicant development efforts, Industry cost share, or future funding (none currently identified).
- Autonomy is a desirable capability; however, autonomy development and maturation are not the focus of this S&T funding.
- System life-cycle affordability has high importance.
- If initial results are promising, a decision may be made to further pursue capability-enhancing ideas.
- For purposes of this Topic, a hybrid aircraft is defined as one that has a propulsive system or architecture that combines multiple elements of energy storage; power production; and or thrust production that may be used simultaneously or separately, either independently or cooperatively. Therefore, supplemental technologies that take advantage of onboard electric power generation, such as electric actuation or advanced mission systems, may be included. Advanced technologies, including fuel cells, may be considered. This definition is provided as a means of conveying intent and openness to different technical solutions. It is not meant to be overly constraining or to hinder innovation. Offerors may propose extensions to this definition for consideration by the Government. Aside from these guidelines, there will not be a government-prescribed, best hybrid system configuration or technology set for this project.
- It is already understood that hybrid technology increases range, payload and mission flexibility compared to all-electric aircraft. Conventional technology (mature, well-known, existing) rotorcraft provide a more relevant comparison baseline.

Goals: The Army desires to define the potential and limitations of hybrid-electric technology as applied to VTOL/STOL aviation to provide positive warfighter outcomes in operational scenarios.

- Leverage Industry development of dual-use technology to enhance the Army's technical knowledge and understanding of hybrid technology for aviation systems in operationally relevant applications.
- Characterize the technical and programmatic risks associated with the transition of various hybrid technologies.
- Demonstrate advancements in Warfighter capabilities and improvements to Army operations.
- Inform the Army's aviation S&T strategy, requirements development, and transition planning.

The following have been deleted:

<u>ATTACH 5 NOFO OR TOPIC</u>

(End of Summary of Changes)